

$$1) \text{ twemple } (4Tf/3 - 4k/3) = \text{twemple} \left(\frac{f - \frac{k}{T}}{\frac{3}{4T}} \right) = x(f) \quad (1)$$

$$R_x(f) = T \text{ rep twemple} \left(\frac{f}{\frac{3}{4T}} \right)$$

$$r_x(kT) = T \cdot \frac{3}{4T} \text{ sinc}^2 \left(\frac{3}{4T} \cdot kT \right) = \frac{3}{4} \text{ sinc}^2 \left(\frac{3}{4} k \right)$$

$3x(kT) - x(kT-T)$ è una v.e. gaussiana con media 0 e varianza

$$\begin{aligned} E [9 x^2(kT) + x^2(kT-T) - 6 x(kT) x(kT-T)] \\ = 9 r_x(0) + r_x(0) - 6 r_x(T) = 10 \cdot \frac{3}{4} - 6 \cdot \frac{3}{4} \text{ sinc}^2 \left(\frac{3}{4} \right) \\ = \sigma^2 \end{aligned}$$

Dunque

$$P [3x(kT) - x(kT-T) \geq 2] = Q \left(\frac{2}{\sigma} \right)$$

$$\begin{aligned} 2) \quad P &= \binom{5}{3} (1-p)^3 p^2 + \binom{5}{4} (1-p)^4 p + \binom{5}{5} (1-p)^5 \\ &= 10 p^2 (1-p)^3 + 5 p (1-p)^4 + (1-p)^5 \end{aligned}$$

$$\begin{aligned} 3) \quad G(f) &= V_0 X(f) \quad X(f) = \mathcal{F} \left[\text{sinc}^4 \left(\frac{t}{T} \right) \right] \text{ la} \\ &\text{estensione spettrale } \left[-\frac{4}{2T}, \frac{4}{2T} \right] \\ &\text{(conv. di 4 tf. con estensione } \left(-\frac{1}{2T}, \frac{1}{2T} \right)) \end{aligned}$$

$$H(f) = h_0 \frac{T}{5} \text{rect} \left(\frac{f}{\frac{5}{T}} \right) \quad \text{estensione} \quad \left(-\frac{5}{2T}, \frac{5}{2T} \right) \quad (2)$$

$$L(f) = A_m \frac{T}{4} \text{rect} \left(\frac{f}{\frac{4}{T}} \right) \quad \text{estensione} \quad \left(-\frac{4}{2T}, \frac{4}{2T} \right)$$

$$G(f) \cdot H(f) \cdot L(f) = G(f) \cdot \frac{h_0 T^2}{20} \cdot A_m = C(f)$$

$$c(f) = \frac{V_0 h_0 T^2 A_m}{20} \text{sinc}^4 \left(\frac{f}{T} \right) \quad \text{è di Nyquist} \\ (\text{si annulla per } t = kT, \quad k \neq 0) \text{ con}$$

$$c(0) = V_0' = \frac{V_0 h_0 T^2 A_m}{20}$$

$$\begin{aligned} \sigma_m^2 &= \int_{-\infty}^{+\infty} R_0 \cdot |H(f)|^2 df = \int_{-\frac{5}{2T}}^{\frac{5}{2T}} R_0 \left(\frac{h_0 T}{5} \right)^2 df \\ &= R_0 \left(\frac{h_0 T}{5} \right)^2 \cdot \frac{5}{T} = \frac{R_0 h_0^2 T}{5} \end{aligned}$$

$$P_e = \frac{10}{6} Q \left(\frac{V_0'}{\sigma} \right)$$

$$h) \quad \Lambda = 10^{\frac{50}{10}} = 10^5 \quad (A_P)_{dB} = 2 \cdot L \quad A_P = 10^{-\frac{2L}{10}}$$

$$\Lambda = \frac{11_T \cdot A_P}{2 R_0 B} \Rightarrow 10^{-\frac{2L}{10}} = \frac{10^5 \cdot 2 \cdot 10^{-18} \cdot 15 \cdot 10^3}{200}$$

$$L = +5 \log_{10} \left(\frac{200}{10^5 \cdot 2 \cdot 10^{-18} \cdot 15 \cdot 10^3} \right) = 54.11 \text{ km}$$

(3)

5) Su $B_c = 320 \text{ kHz}$ si può trasmettere $\frac{1}{T} = 320 \cdot 10^3 \text{ simboli/s}$,
ovvero, con una 4-QAM, $640 \cdot 10^3 \text{ bit/s}$

Per un segnale con banda $B = 20 \text{ kHz}$, occorre
prelevare $40 \cdot 10^3 \text{ campioni/s}$. Ciascun campione
può dunque essere rappresentato con

$$\frac{640 \cdot 10^3}{40 \cdot 10^3} = 16 \text{ bit}$$

Il rapporto segnale/minore di quantizzazione è

$$A = L^2 = 2^{32} \quad (A)_{dB} \approx 96 = 16 \cdot 6 \text{ dB}$$